

Lecture 5 - Chemical potential and the Numerical Sign Problem

Simran Singh

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Abstract

In this lecture we look at what happens when we try to study QCD at finite baryon density on the lattice. We start by seeing how a chemical potential μ is introduced into the lattice theory. We then show why working at finite density causes the fermion determinant to become complex – breaking a property called γ^5 -hermiticity – and why this leads to the *numerical sign problem*, one of the biggest unsolved challenges in lattice QCD. We finish by discussing what this means for our ability to study the QCD phase diagram computationally.

1 Overview

The QCD phase diagram is a map of how strongly interacting matter – made up of quarks and gluons – behaves with varying temperatures T and baryon densities, parametrised by the chemical potential μ . Getting a good understanding of this phase structure matters because: it tells us about the interior of neutron stars, and helps us understand what happens to matter in heavy-ion collision experiments, where extreme temperatures and densities are reached for a very brief moment.

The problem is that QCD is a strongly coupled theory across much of this phase diagram, so perturbation theory – which works so well in particle physics at high energies – is not very useful here. This is exactly the kind of situation where lattice QCD would in principle shine: it is a non-perturbative, first-principles numerical approach that does not rely on the coupling being small. It is therefore natural to ask: *can lattice QCD give us the full QCD phase diagram?*

As we will see in this lecture, the answer is unfortunately *no*. The standard way to perform lattice QCD calculations – Monte Carlo importance sampling – relies on the path integral weight e^{-S} being a real, positive number so that it can be interpreted as a probability. At finite baryon density, this breaks down: the fermion determinant becomes complex, and the weight is no longer positive. This is the origin of the *numerical sign problem*, and it makes Monte Carlo sampling exponentially hard.

The conjectured QCD phase diagram is shown in Fig. 1. Because of the sign problem, conventional lattice calculations are only reliable near zero chemical potential, $\mu \approx 0$. In this region, lattice studies have shown that as you cool the quark-gluon plasma down, it transitions smoothly into ordinary hadronic matter – this is called a *crossover* [2]. At the other extreme, at very high baryon densities, model-based and perturbative calculations suggest that the transition becomes *first order* [3]. If

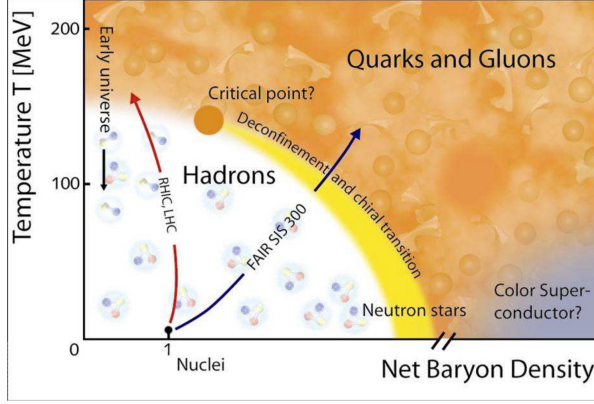


Figure 1: Conjectured QCD phase diagram. [Image from: [1]]

both of these results are correct, then somewhere in the middle there must be a point where the crossover ends and the first-order line begins – this is called the *second-order critical endpoint*. Finding this endpoint is one of the key goals of current heavy-ion experiments, but since direct lattice calculations at finite μ are not yet possible, all existing predictions for its location rely on indirect methods and cannot be fully trusted.

2 Introduction to the Chemical Potential

The chemical potential becomes important the moment we ask questions about the *density* of a system rather than just its temperature: what is the average particle number, how does it respond to changes in temperature and external conditions, and how large is the excess of matter over antimatter?

Formally the chemical potential is the thermodynamic variable conjugate to the conserved charge – in QCD this is the baryon number – and it enters the **grand-canonical** partition function as,

$$Z_{GC}(T, \mu) = \text{Tr} \left[e^{-(\hat{H} - \mu_B \hat{N}_B)/T} \right]. \quad (1)$$

The baryon number operator is related to the quark number operator $\hat{N}_B = \frac{\hat{N}_q}{3}$ and the baryon chemical potential is related to the quark chemical potential as $\mu_B = 3\mu_q$. The average particle number can be obtained from Eq. 1 as,

$$n_q \equiv \frac{1}{V} \langle \hat{N}_q \rangle = \frac{T}{V} \frac{\partial \ln Z(T, \mu)}{\partial \mu_q}, \quad (2)$$

where n_q is referred to as the *number density*. From now on we will drop the subscript q on the chemical potential and number operator to mean the quark quantities implicitly.

2.1 Treatment in the continuum

Since we are interested in QCD, we will use the language developed in Lecture 3 (Lecture 2 in the notes) for the fermions. The conserved current associated to the

Dirac lagrangian is,

$$j_\mu(x) = \bar{\psi}(x)\gamma_\mu\psi(x), \quad (3)$$

where the conserved charge – here the quark number – is given by the space integration of the 4th component of the conserved current in Eq. 4 as,

$$N = \int d^3x \bar{\psi}(x)\gamma_4\psi(x). \quad (4)$$

Hence, one is tempted to incorporate into the Dirac lagrangian, a term of the form $\mu\bar{\psi}(x)\gamma_4\psi(x)$. It was shown in [4], that adding the chemical potential term in this way *on the lattice* leads to divergent quantities in the continuum. In order to see this, consider the energy density given by (looking at Eq. 1, with V being the spatial volume),

$$\epsilon(\mu) = \frac{1}{V} \langle \hat{H} - \mu \hat{N} \rangle = -\frac{1}{V} \frac{\partial}{\partial \beta} \ln Z \Big|_\mu. \quad (5)$$

In the exercise you will show that this quantity, leads to divergence of the form,

$$\lim_{a \rightarrow 0} (\epsilon(\mu) - \epsilon(0)) \propto \frac{\mu^2}{a^2}, \quad (6)$$

if a chemical potential is added in the form of Eq. 4.

2.2 Treatment on the lattice

Recall the naive fermion action – now with an external gauge field,

$$S_F[\psi, \bar{\psi}, U] = a^4 \sum_n \bar{\psi}(n) \left(\sum_{\mu=1}^4 \gamma_\mu \frac{U_\mu(n)\psi(n+\hat{\mu}) - U_{-\mu}(n)\psi(n-\hat{\mu})}{2a} + m\psi(n) \right). \quad (7)$$

As shown in Section 2.1, naively adding a $\mu\bar{\psi}\gamma_4\psi$ term to this action leads to quadratic divergences in the continuum limit. The correct approach, due to Hasenfratz and Karsch [4], is to introduce the chemical potential by modifying the *temporal hopping terms*. The motivation is that, in the Euclidean grand-canonical formalism

$$Z = \text{Tr} e^{-(\hat{H} - \mu \hat{N})/T}, \quad (8)$$

the chemical potential enters as the fourth (temporal) component of an imaginary, constant abelian gauge potential, $A_4 = -i\mu$. Since gauge fields couple to lattice fermions through the link variables as phases e^{iaA_4} on the hopping terms – it is natural to introduce μ in the same way, by multiplying the forward and backward temporal hopping terms by some functions $f(\mu a)$ and $f(\mu a)^{-1}$, respectively (we will argue what these functions should be). This minimal-coupling structure is precisely what removes the quadratic divergence of the naive prescription.

Concretely, the temporal part of the fermion action is modified according to

$$\bar{\psi}(n) \gamma_4 \frac{f(\mu a) U_4(n) \psi(n+\hat{4}) - f(\mu a)^{-1} U_4(n-\hat{4})^\dagger \psi(n-\hat{4})}{2a}, \quad (9)$$

where we have used $U_{-\nu}(n) = U_\nu(n-\hat{\nu})^\dagger$. Notice that the forward temporal link picks up a factor $f(\mu a)$ while the backward link picks up its inverse $f(\mu a)^{-1}$; this asymmetry is essential for recovering the correct continuum limit – due to the way the links appear in the Eq. 9. The function $f(\mu a)$ must satisfy the following properties::

Recall that we are in Euclidean space, so the zeroth component of Minkowski spacetime corresponds to the 4th component in Euclidean.

A guided derivation of this will be an exercise.

It is important to note at this stage that in the next part of the lecture when you study the Hamiltonian lattice formulation – this problem does not arise – you can simply add the term $\mu\hat{N}$ to your Hamiltonian. The reason is that time is *not* discretized in the Hamiltonian formulation.

- **Correct $\mu = 0$ limit.** At vanishing chemical potential the original action must be recovered:

$$f(0) = 1. \quad (10)$$

- **Correct continuum density term.** Expanding to first order in a must reproduce the continuum coupling $\mu \bar{\psi} \gamma_4 \psi$, which fixes the slope at the origin:

$$f(\mu a) = 1 + \mu a + \mathcal{O}((\mu a)^2) \iff f'(0) = 1. \quad (11)$$

- **Antisymmetry under $\mu \rightarrow -\mu$.** Reversing the sign of the chemical potential exchanges particles and antiparticles, and hence the roles of the forward and backward temporal links. This requires

$$f(\mu a) = \frac{1}{f(-\mu a)}. \quad (12)$$

The exponential $f(\mu a) = e^{\mu a}$ follows directly from the gauge-field analogy: it is simply the link phase e^{iaA_4} evaluated at $A_4 = -i\mu$. One readily checks that it satisfies all three conditions above:

$$f(0) = 1, \quad f'(0) = 1, \quad f(\mu a) = \frac{1}{f(-\mu a)}. \quad (13)$$

In retrospect this is a very natural result: a quark hopping one step in Euclidean time picks up the Boltzmann factor $e^{\mu a}$, so propagating across the full temporal extent N_τ accumulates $e^{\mu a N_\tau} = e^{\mu/T}$, since $a = 1/(N_\tau T)$. Substituting back, the modified temporal hopping term reads

Recall how temperature was introduced in the last lecture by Alessio.

$$\bar{\psi}(n) \gamma_4 \frac{e^{\mu a} U_4(n) \psi(n + \hat{4}) - e^{-\mu a} U_4(n - \hat{4})^\dagger \psi(n - \hat{4})}{2a}, \quad (14)$$

while the spatial hopping terms remain unchanged. The full fermion action $S_F[\bar{\psi}, \psi, U, \mu]$ is obtained by replacing only the $\mu = 4$ terms in the original action with the expression above.

3 Computing observables via Monte Carlo

The full QCD partition function including both gauge fields and fermion fields can be written as,

$$Z = \int \mathcal{D}U \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{-S_G[U] - S_F[\bar{\psi}, \psi, U, \mu]}, \quad (15)$$

where $S_G[U]$ is the pure gauge (Wilson) action and $S_F[\bar{\psi}, \psi, U, \mu]$ is the fermion action with chemical potential as derived in the previous section. We saw in the previous lecture that the fermion action is not a "number" point wise and we must integrate out the Grassmann fields in order to be able to compute numerical quantities. The Grassmann integration over the fermion fields can then be performed exactly using the standard result for Gaussian Grassmann integrals – as you already showed in the exercise sheets,

$$\int \mathcal{D}\bar{\psi} \mathcal{D}\psi e^{-\bar{\psi} M \psi} = \det M(U), \quad (16)$$

leaving us with an effective partition function in terms of the gauge links only,

$$Z = \int \mathcal{D}U \det M(U) e^{-S_G[U]}. \quad (17)$$

In order to evaluate $\langle \mathcal{O} \rangle$ numerically, we interpret the combination $\det M(U) e^{-S_G[U]}$ as a probability weight and draw a representative set of gauge configurations $\{U^{(1)}, U^{(2)}, \dots, U^{(N)}\}$ distributed according to this weight. The expectation value is then estimated as a simple average over the ensemble,

$$\langle \mathcal{O} \rangle \approx \frac{1}{N} \sum_{i=1}^N \mathcal{O}[U^{(i)}]. \quad (18)$$

This procedure is called *importance sampling*: instead of sampling configurations uniformly (which would be hopelessly inefficient given the enormous configuration space), we concentrate sampling in the regions of configuration space that contribute most to the path integral. This is the standard Monte Carlo approach used in virtually all lattice QCD simulations.

The crucial requirement for this to work is that the weight,

$$W[U] = \det M(U) e^{-S_G[U]}, \quad (19)$$

must be *real and positive* so that it can genuinely be interpreted as a probability distribution. The factor $e^{-S_G[U]}$ is always real and positive by construction. As we have also seen in the previous lectures, due to the γ^5 -hermiticity of the Dirac operator,

$$\gamma^5 M(U) = M(U)^\dagger \gamma^5, \quad (20)$$

we obtained $\det M(U) = [\det M(U)]^*$ – ensuring positivity via complex conjugate eigenpairs. Whether the modification of the fermion matrix in the presence of chemical potential $\det M(U, \mu)$ is real and positive is a separate question – and as we will see in the next section, this is precisely where things go wrong at finite μ .

4 Loss of γ^5 hermiticity of the fermion matrix

Recall from Lecture 3 that the γ^5 -hermiticity of the Euclidean Dirac operator $D \equiv \gamma_\mu^E \partial_\mu + m$ was the key property that guaranteed $\det D$ is real and positive. Let us briefly recall why it holds, and then show explicitly that introducing a chemical potential μ breaks it.

Review: γ^5 -hermiticity at $\mu = 0$

The Euclidean γ -matrices satisfy $(\gamma_\mu^E)^\dagger = \gamma_\mu^E$ (they are Hermitian), and anti-commute with γ^5 : $\{\gamma^5, \gamma_\mu^E\} = 0$. Starting from the fermion matrix $M = \gamma_\mu^E \partial_\mu + m$, we act with γ^5 from the left,

$$\begin{aligned} \gamma^5 M &= \gamma^5 (\gamma_\mu^E \partial_\mu + m) \\ &= (-\gamma_\mu^E \partial_\mu + m) \gamma^5, \end{aligned} \quad (21)$$

Note change of notation from $D \rightarrow M - M[U]$ is the standard notation in lattice literature.

where we used $\gamma^5 \gamma_\mu^E = -\gamma_\mu^E \gamma^5$. On the other hand, taking the Hermitian conjugate of M and recalling that $\partial_\mu^\dagger = -\partial_\mu$ (anti-Hermitian) and $(\gamma_\mu^E)^\dagger = \gamma_\mu^E$,

$$M^\dagger = (\gamma_\mu^E)^\dagger \partial_\mu^\dagger + m = -\gamma_\mu^E \partial_\mu + m. \quad (22)$$

Comparing, we see that $\gamma^5 M = M^\dagger \gamma^5$, i.e. the fermion matrix is γ^5 -Hermitian. This is what ensures that eigenvalues of M come in complex conjugate pairs and $\det M$ is real and positive.

Breaking of γ^5 -hermiticity at $\mu \neq 0$

Now consider the fermion matrix with a chemical potential, $M[U, \mu]$. As we discussed in Section 2, the chemical potential enters by modifying the temporal hopping terms, which at the level of the continuum-like operator amounts to the replacement,

$$M[U, \mu] = \gamma_\mu^E \partial_\mu + m + \mu \gamma_4^E. \quad (23)$$

We now repeat the same exercise. Acting with γ^5 from the left and using $\{\gamma^5, \gamma_\mu^E\} = 0$,

$$\begin{aligned} \gamma^5 M[U, \mu] &= \gamma^5 (\gamma_\mu^E \partial_\mu + m + \mu \gamma_4^E) \\ &= (-\gamma_\mu^E \partial_\mu + m - \mu \gamma_4^E) \gamma^5. \end{aligned} \quad (24)$$

Taking the Hermitian conjugate of $M[U, \mu]$, and using $(\gamma_4^E)^\dagger = \gamma_4^E$ (with μ real for now),

$$M^\dagger[U, \mu] = -\gamma_\mu^E \partial_\mu + m + \mu \gamma_4^E. \quad (25)$$

Comparing the two expressions, we see that $\gamma^5 M[U, \mu] \neq M^\dagger[U, \mu] \gamma^5$ – the $\mu \gamma_4^E$ term has opposite sign. Instead, we find,

$$\gamma^5 M[U, \mu] = M^\dagger[U, -\mu] \gamma^5. \quad (26)$$

γ^5 -hermiticity is *broken* at finite real μ : the fermion matrix at chemical potential μ is related by γ^5 conjugation not to its own Hermitian conjugate, but to the Hermitian conjugate of the matrix at *opposite* chemical potential $-\mu$. Taking the determinant of Eq. (26) and using $\det M^\dagger = (\det M)^*$,

$$\det M[U, \mu] = \det M^\dagger[U, -\mu] = (\det M[U, -\mu])^*. \quad (27)$$

Crucially, $M[U, \mu]$ and $M[U, -\mu]$ are *different* matrices, so $\det M[U, \mu]$ is related to the complex conjugate of *another* determinant, not its own. There is therefore no reason for $\det M[U, \mu]$ to be real, and in general it is *complex* at finite real μ : **this is the root cause of the sign problem in lattice QCD.**

So far μ has been real, which was all we needed to expose the sign problem. It is worth keeping μ *complex* for a moment, as this extra generality is what underlies simulations at imaginary chemical potential. With μ complex, the Hermitian conjugate carries μ^* rather than μ , since $(\mu \gamma_4^E)^\dagger = \mu^* \gamma_4^E$:

$$M^\dagger[U, \mu] = -\gamma_\mu^E \partial_\mu + m + \mu^* \gamma_4^E. \quad (28)$$

Repeating the comparison with $\gamma^5 M[U, \mu]$ then gives

$$\gamma^5 M[U, \mu] = M^\dagger[U, -\mu^*] \gamma^5, \quad (29)$$

which reduces to Eq. (26) for real μ . Taking the determinant,

$$(\det M[U, \mu])^* = \det M[U, -\mu^*]. \quad (30)$$

For real μ this collapses back to Eq. (27). But it also covers the case of *purely imaginary* chemical potential, $\mu = i\mu_I$ with $\mu_I \in \mathbb{R}$: there $-\mu^* = -(-i\mu_I) = i\mu_I = \mu$, so $(\det M[U, \mu])^* = \det M[U, \mu]$ and the determinant is *real*. This is the basis of lattice simulations at imaginary μ , which are free of the sign problem and can be analytically continued back to real μ .

5 Complex action problem

In order to understand the damage caused by the non-reality of $\det M[U, \mu]$, consider again the partition function (now dependent on the chemical potential μ),

$$Z(\mu) = \int \mathcal{D}U \det M(U, \mu) e^{-S_G[U]}. \quad (31)$$

Recall that Monte Carlo simulations are performed under the assumption that the integrand of the path integral is a well defined probability measure,

$$p(U, \mu) = \det M(U, \mu) e^{-S_G[U]}, \quad (32)$$

according to which one generates configurations U , on which one computes observables $O(U)$ and computes averages according to Eq. 18. Hence, if $p(U)$ is not a probability but a complex number – what are we supposed to do? There are a few obvious choices – each with its own limitation. We will consider two of the most common choices below: Phase quenching and sign quenching.

Reduction to Numerical Sign Problem:

The first natural argument to make is that the grand canonical partition function is a physical quantity which has a perfectly valid interpretation as a normalisation constant for a system at finite chemical potential. Looking at the integrand of Eq. 31, the implication for $Z(\mu)$ to be real is that we only need to consider the real part of the probability weight: $\text{Re}(\det M(U, \mu))$,

$$Z(\mu) = \int \mathcal{D}U \text{Re}(\det M(U, \mu)) e^{-S_G[U]}. \quad (33)$$

An immediate concern should be that this does not guarantee *positivity* of the probability weight. We must further take the absolute value of this quantity to actually be able to compute physical observables,

$$Z(\mu)_{\text{pos}} = \int \mathcal{D}U |\text{Re}(\det M(U, \mu))| e^{-S_G[U]}. \quad (34)$$

At the end we must correct for the signs when going from $Z(\mu)_{\text{pos}} \rightarrow Z(\mu)$ and with the help of a toy example we will see that this leads to a signal-to-noise problem.

Phase Quenching and the Overlap problem

Another quantity to consider as a valid weight factor is the absolute value of the probability weight,

$$Z(\mu)_{\text{phase quenched}} = Z_{\text{pq}}(\mu) = \int \mathcal{D}U |\det M(U, \mu)| e^{-S_G[U]}. \quad (35)$$

This should be contrasted with the full theory as given in Eq. 31, and observables in both theories are related by,

$$\begin{aligned} O(\mu)_{\text{full}} &= \frac{\int \mathcal{D}U O(U) |\det M(U, \mu)| e^{i\varphi} e^{-S_G[U]}}{\int \mathcal{D}U |\det M(U, \mu)| e^{i\varphi} e^{-S_G[U]}} \\ &= \frac{\int \mathcal{D}U O(U) |\det M(U, \mu)| e^{i\varphi} e^{-S_G[U]} / Z(\mu)_{\text{pq}}}{\int \mathcal{D}U |\det M(U, \mu)| e^{i\varphi} e^{-S_G[U]} / Z(\mu)_{\text{pq}}} = \frac{\langle O e^{i\varphi} \rangle_{\text{pq}}}{\langle e^{i\varphi} \rangle_{\text{pq}}}. \end{aligned} \quad (36)$$

where $\langle \cdot \rangle_{\text{pq}}$ denotes expectation values taken with respect to the phase-quenched measure with weight $|\det M(U, \mu)| e^{-S_G[U]}$, normalised appropriately by $Z(\mu)_{\text{pq}}$. The quantity in the denominator $\langle e^{i\varphi} \rangle$ is called the *average phase* and quantifies how bad the sign problem in your theory is. It scales exponentially with volume,

$$\langle e^{i\varphi} \rangle = \frac{Z(\mu)_{\text{full}}}{Z(\mu)_{\text{pq}}} = e^{-V\Delta f}, \quad (37)$$

where $\Delta f > 0$ is the difference in the intrinsic free energies of the two theories Z_{full} and $Z_{\text{phase quenched}}$. Hence if the two theories don't have sufficient overlap, this average phase approaches zero and we are unable to use Eq. 36 to compute anything because it becomes undefined.

5.1 Practical example: Toy example

To make the problems described above explicit and concrete, we consider a simple toy model: a 0-dimensional field theory (no spacetime at all) with partition function,

$$Z_{\text{toy}}(\mu) = \int_{-\infty}^{\infty} d\phi e^{-\phi^2 + i\mu\phi}. \quad (38)$$

Here ϕ plays the role of the field and μ plays the role of the chemical potential. There is no spacetime volume in this model, so the exponential volume-worsening of the sign problem instead manifests as exponential worsening in μ . Despite its simplicity, this model captures both the overlap problem and the sign problem in a transparent way.

The overlap problem

The integral $Z_{\text{toy}}(\mu)$ can be computed exactly by completing the square in the exponent,

$$-\phi^2 + i\mu\phi = -\left(\phi - \frac{i\mu}{2}\right)^2 - \frac{\mu^2}{4}, \quad (39)$$

which gives $Z_{\text{toy}}(\mu) = \sqrt{\pi} e^{-\mu^2/4}$. The phase-quenched partition function is simply $Z_{\text{toy}}(0) = \sqrt{\pi}$, so the average phase is,

$$\langle e^{i\varphi} \rangle = \frac{Z_{\text{toy}}(\mu)}{Z_{\text{toy}}(0)} = e^{-\mu^2/4}. \quad (40)$$

This shows clearly how the overlap between the full and phase-quenched theories decreases exponentially as μ increases. In the full QCD case, this exponent would be $e^{-V\Delta f}$, so here $\mu^2/4$ plays the role of $V\Delta f$. For large μ , the average phase $\langle e^{i\varphi} \rangle \rightarrow 0$, and the reweighting formula Eq. (36) becomes 0/0 – numerically useless. This is illustrated in Fig. 3, which shows how the phase-quenched Gaussian and the full oscillating integrand become increasingly misaligned as μ grows.

The sign problem

To see the sign problem directly, we write the integrand of $Z_{\text{toy}}(\mu)$ in terms of its real part,

$$\text{Re} \left[e^{-\phi^2 + i\mu\phi} \right] = e^{-\phi^2} \cos(\mu\phi). \quad (41)$$

For $\mu = 0$ this is just a Gaussian, which is everywhere positive and easy to sample. As μ increases, the $\cos(\mu\phi)$ factor introduces rapid oscillations: the integrand alternates sign over a length scale $\sim 1/\mu$ in ϕ -space, while the Gaussian envelope $e^{-\phi^2}$ decays over a length scale of order 1. To get the integral, one must accurately compute the near-cancellation between large positive and negative contributions – the signal is the tiny difference between nearly equal positive and negative areas. In a Monte Carlo simulation, this cancellation must be resolved from noisy samples, and since the noise does not cancel but the signal does, the signal-to-noise ratio deteriorates rapidly. This is illustrated in Fig. 2: for large μ the integrand oscillates so rapidly that straightforward Monte Carlo is unable to resolve the integral.

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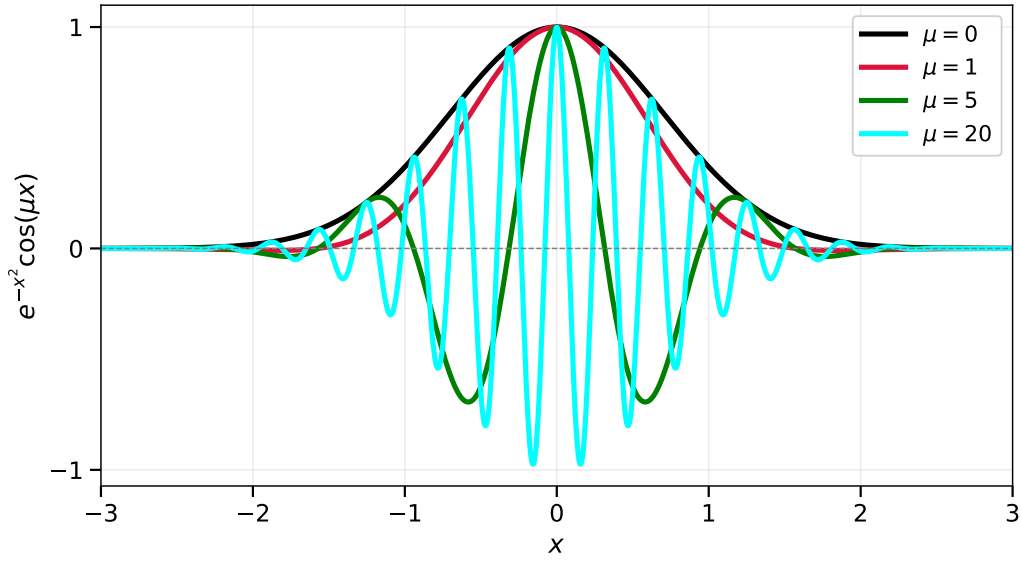


Figure 2: Sign problem via the toy model: the real part of the integrand $e^{-\phi^2} \cos(\mu\phi)$ is shown for increasing values of μ . As μ grows, rapid oscillations make Monte Carlo estimation of the integral exponentially hard.

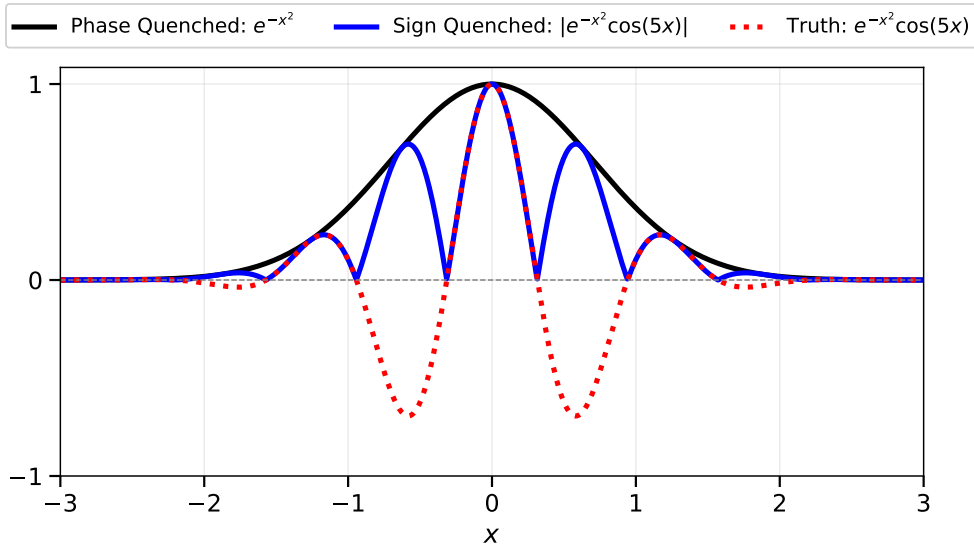


Figure 3: Overlap vs signal-to-noise problem via the toy model: the phase-quenched weight $e^{-\phi^2}$ (black solid line) is compared to the full integrand for increasing μ . The two distributions have decreasing overlap, making re-weighting from the phase-quenched ensemble exponentially costly. In contrast, the sign quenched curve (blue solid line) and the full integrand (red dotted line) have good overlap.